

Shubham Manoj Heda

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EDUCATION

Master of Science in Information Systems, University of Maryland, Baltimore County	Aug 2019 - May 2021
Bachelor of Engineering in Information Technology, Pune University	Aug 2014 - Jun 2018

TECHNICAL SKILLS

Programming Languages: Python 3, SQL, PL/SQL, R, UNIX, C | **Cloud:** AWS, GCP | **ETL:** Airbyte, Talend, SSIS
Database Management: Snowflake, BigQuery, MS SQL Server, MySQL, PostgreSQL, Oracle 11g and Mongo DB
Knowledge Area: ETL, Data Warehousing, Data Modeling, CI/CD | **Visualization:** Power BI, Tableau, Looker, Excel
Software & Tools: dbt, Apache Airflow, Terraform, Git, Docker, Smartsheet, SharePoint, LucidChart, Cdata, JIRA, Sage, Zendesk, Salesforce | **Libraries:** Pandas, NumPy, SciPy, Matplotlib, PySpark, Scikit-learn
Soft Skills: Leadership, Problem-Solving, Critical Thinking, Attention to Detail, Collaboration, Prioritization

WORK EXPERIENCE

ICAST | Data Engineer | CO, USA Aug 2021 - Feb 2025

- Led a **cross-functional** team to extract and integrate **3.5M+** rows of financial and operational data from **Sage Intacct** into **CData Cloud Connect** resulting in automating workflows and reducing **manual work by 23%**. Used **SQL** queries to validate data accuracy, identifying and resolving discrepancies that improved data integrity by **47%**.
- Designed **data models** in Snowflake based on business requirements and developed optimized **schema architectures** to improve query performance and ensure **data accuracy of 96%**.
- Extracted **financial data** (GL, Projects, AP, AR) from **Cdata Cloud Connect** into **AWS S3** using **PySpark in Amazon EMR**, and streamlined processing time by **9 minutes** before loading into **Snowflake via Snowpipe**.
- Developed an **ETL pipeline** to extract structured data from **Smartsheet API** and load it into **Snowflake** using **Python (pandas, Snowflake Connector, Smartsheet SDK)** and orchestrated the ETL process using **Apache Airflow** running on **AWS EC2** instance.
- Facilitated Snowflake data processing by implementing **Streams & Tasks** for incremental loads, reducing **batch processing by 6 minutes**, and utilizing **Time Travel** for replications.
- Significantly optimized the processing time of Snowflake by creating **Materialized Views** and **Warehouse tuning**. Utilized **Zero-Copy Cloning** to create instant dataset replicas for testing and development. Enforced **RBAC (Role-Based Access Control)** to manage permissions, ensuring secure, role-based access to sensitive data.
- Transformed raw data with **dbt**, improving query performance, and fashioned **dbt macros** and **Jinja** functions to improve code reusability and reduce manual efforts by **42%**. Implemented **dbt tests** to enforce data validation and improve data reliability.
- Created a **CI/CD pipeline** using Git and dbt automating **data model testing, version control, and deployment** resulting in enhanced deployment efficiency, seamless updates to production, and reliable data transformation.
- Cultivated **Power BI dashboards** for departments like **Sales, Marketing, and Construction** by extracting datasets from **Snowflake** using native connectors. Created Financial Reports like **revenue and gross margin** for the **Accounting** team, which helped significantly improve the company's **financial health, performance, and decision-making**.
- Conceptualized and implemented **DAX measures**, reducing report generation time by **40% (from 12 minutes to 7 minutes)** and reducing refresh time by **6 mins**.

Zensar Technologies | Data Analyst | India Jan 2017 – Jul 2019

- Developed automated data pipelines from external data sources (web pages, API, etc.) to the internal data warehouse (**SQL Server, BigQuery**) reducing manual integration time by **42%**. Cleaned and preprocessed data using **Python dataframes** and exported the data to **Tableau** for visualization.
- Utilized Python packages (**NumPy, Pandas, Matplotlib, SciPy, Scikit-learn**) for feature engineering and built machine learning models and boosting model accuracy by **3 times**, and increased prediction reliability for **key business metrics**.
- Designed **Entity-Relationship Diagrams (ERDs)** using **LucidChart** and constructed an **SQL** database to store and manage **53K** IoT device data, ensuring seamless data storage and real-time analytics.
- Created advanced Tableau visualizations (**Scatter Plots, Histograms, Heat Maps**) to represent complex data insights, resulting in a **notable increase in actionable insights** leading to **informed decision-making**.
- Constructed Tableau reports and interactive dashboards, enhancing data accessibility for stakeholders and enabling the Sales team to optimize strategies, resulting in a **2X increase** in sales performance within **8 months**.
- Implemented **Agile** methodology, including daily Scrums and bi-weekly sprint planning, leading to a **25% reduction in project delivery time** and enhancing **cross-team collaboration by 29%**.